

Question 1:-

1. Take a number N and
Print numbers from N down to 1:

Algorithm

1. Start
2. Input N .
3. set $i = N$
4. while $i \geq 1$
 - Print i
 - subtract 1 from i
($i = i - 1$)
5. End.

Pseudocode :-

```

Begin
    Read  $N$ 
    For current( $i$ ) From  $N$  down to 1.
        Do Print current.
    End For
End.
```

Question 2:-

Input a year and determine whether it is a leap year
(Rule: divisible by 4 and (not divisible by 100 OR divisible by 400)).

Algorithm:-

1. Start
2. Input Year
3. $\text{if } (\text{Year} \% 4 == 0 \text{ AND } (\text{Year} \% 100 \neq 0 \text{ OR } (\text{Year} \% 400 == 0)))$
 - Print "Leap Year".
 - Else:
 - Print not a leap year.
4. End.

Pseudocode.

Begin

Read year

$\text{if } (\text{Year} \% 4 == 0 \text{ and } (\text{Year} \% 100 \neq 0 \text{ OR } (\text{Year} \% 400 == 0)))$ then
Print Leap Year

Else

Print not a leap year

End if

End.

Date: _____

Question # 3 : Take a number, count digits and state single / double / tripple-

Algorithm:-

- 1/ Start
- 2/ input num & store it
- 3/ Set count = 0
- 4/ Set temp = num
- 5/ if num == 0, then
Set count = 1 and goto step 8.
- 6/ while temp > 0:
temp = temp / 10 (remove Last digit).
Count = count + 1
- 7/ End while
- 8/ Print count.
- 9/ if count == 1, print "single digit"
Else if count == 2, print "double digit"
Else if count >= 3, Print "trippel digit"
or more"
- 10/ Stop.

Pseudocode:-

Date: _____

Begin

input num

set count = 0.

set temp = num.

if num == 0 then

set count = 1

Else.

while temp > 0 D

Set temp = temp / 10

Set count = count + 1

End while

End if

Print "total digits:" , Count.

if count == 1 then

Print "single digit"

Else if count == 2 then

Print "double digit"

Else.

Print "triple digit or more"

End if

End.

Question 4:-

Print the multiplication-tables from 1 to 10, but skip the table of any no. that is divisible by 3.

Algorithm

- 1, Start
- 2, For $i = 1$ to 10 DO
- 3, If $i \bmod 3 \neq 0$ then
- 4, Continue to next i
[skip this table]
- 5, End if
- 6, Print "table of ", i
- 7, For $j = 1$ to 10 DO
- 8, result = $i * j$
- 9, Print i , "x", j = " , result.
- 10, End for
- 11, End for
- 12, End.

Pseudocode :-

Begin

For $i \leftarrow 1$ to 10

 If $i \% 3 == 0$ then

 Continue

 End if

 Print "multiplication table of" + i

 For $j \leftarrow 1$ to 10

 Print $i + "*" + j + "=" + (i*j)$

 End for

End for

End.

Question #5

take a number, repeatedly sum its digits until the result is a single digit, and displays every intermediate result.

Algorithms:-

- 1/ Start
- 2/ input number num
- 3/ if $\text{num} < 0$ then $\text{num} = -\text{num}$
- 4/ Print "Original number:", num.
- 5/ While $\text{num} \geq 10$ Do
- 6/ $\text{temp} = \text{num}$
- 7/ $\text{sum} = 0$
- 8/ while $\text{temp} > 0$ Do
- 9/ $\text{digit} = \text{temp} \bmod 10$
- 10/ $\text{sum} = \text{sum} + \text{digit}$
- 11/ $\text{temp} = \text{temp} \div 10$
- 12/ End while
- 13/ Print "sum=", sum
- 14/ $\text{num} = \text{sum}$
- 15/ End while
- 16/ Print "final single digit:", num
- 17/ End-

Pseudocode :-

Date:

Begin

Read num

num $\leftarrow$ ABS(num).

if num < 10 then

Print "Already single digit:" + num

Else.

while num ≥ 10

temp $\leftarrow$ num

sum $\leftarrow 0$

while temp > 0

-- digit $\leftarrow$ temp % 10

--- sum $\leftarrow$ temp / 10

// integer division

End while

Print sum

num $\leftarrow$ sum

End while

Print "digital root is:" + num

End if

End.

Question 1:-
Hotel booking system.

Algorithm

Start

input season, , room type, nights.
check season

if season == Peak then

if room type == standard then
rate = 5000

Else if deluxe then rate = 8000

Else rate = 12000

Else. (off-peak).

if room type == standard then
rate = 3000

Else if ^{deluxe then} rate = 8000

Else rate = 8000.

total = rate * nights.

if nights > 7 then

discount = total * 0.15,

Final price = total - discount

Else final price = total

total revenue = total revenue + final price

display final price, total revenue

End if

End.

Hotel booking System.

PAC

Given data

- N guests.
- * Season (Peak/off-peak)
- * Room type (standard, deluxe, suite).
- * nights stayed.
- * Peak rates = Rs 5000/night ^{standard.}
- * Deluxe = Rs 8000/night.
- * Suit = Rs. 12000/night
- * Off- Peak rates.
- * nights > 7 = 15% disc.

Required result.

final price of each guests (after discount if applicable)
Total total revenue (sum of all guests) displayed.
after Loop ends.

Processing Required.

input N

Loop for N guests.

Nested if = check season & room type to select rate.

total = rate * nights

if nights > 7 then discount = total

10-15 else 0.

final price = total - discount

revenue += final price

display final price

after loop display revenue.

Solution alternatives

use nested if-else for season and room type.

Use switch-case for room type inside if.

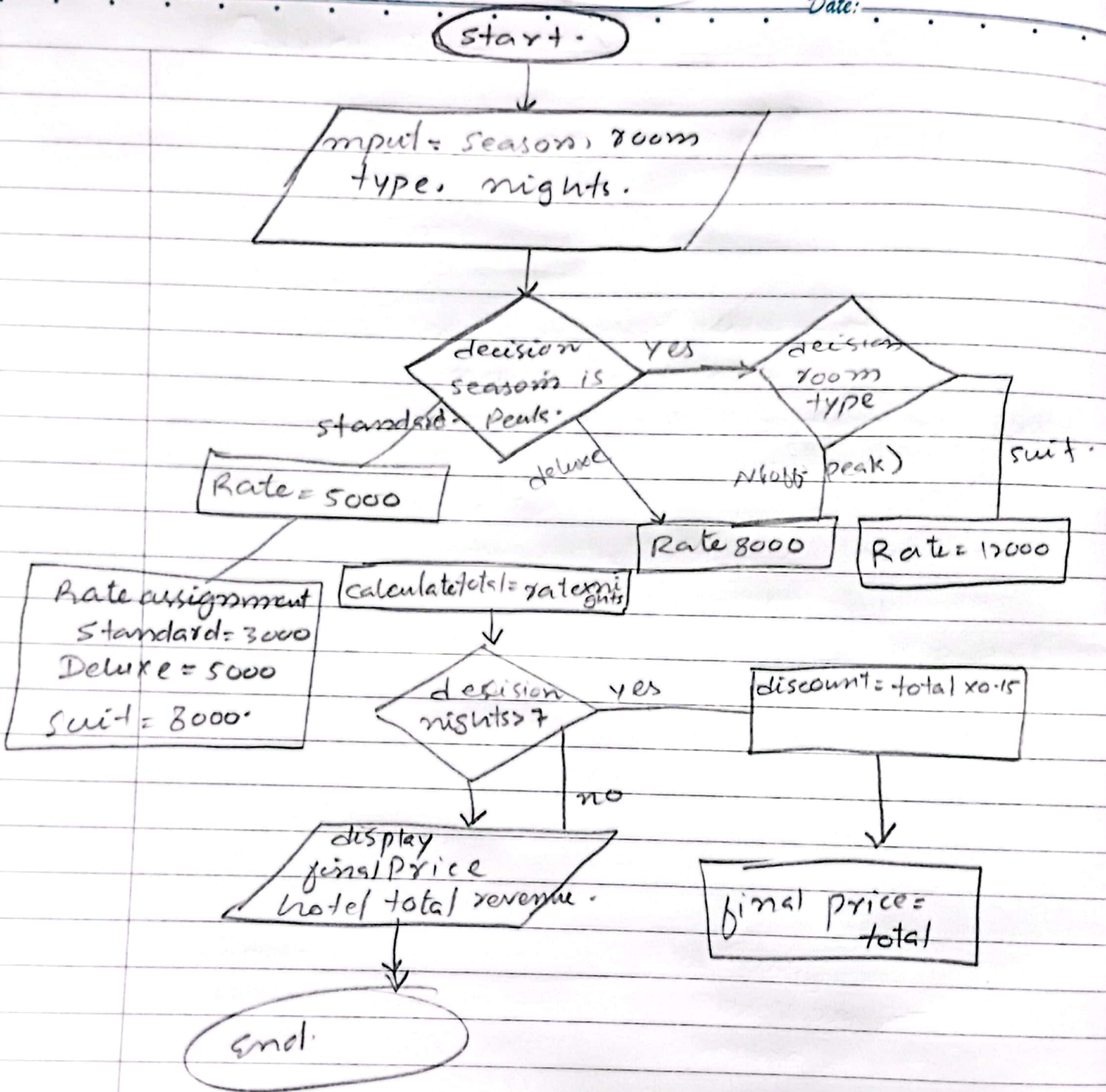
Use 2D array/look up table for rates to avoid nesting.

3 IPO chart

input	processing	output
*Season (string)	* determine rate based on season & room type.	rate
*room type (string)	* $total = rate * nights$	total bill
*nights (int)	* if nights > 7 apply 15% discount	discount
	* final price = total - discount	final price
	* hotel revenue + final price	hotel total revenue

Flow chart.

Date:



Pseudocode:

Date:

Begin

Read season, Room type, nights,
total revenue = 0

if season == "Peak" then

if room type == "standard" then
rate = 5000.

Else if room type == "deluxe" then rate = 8000.

Else rate = 12000.

Else.

if room type == "standard" then rate = 3000

Else if room type == "deluxe" then rate = 5000.

Else rate = 8000

End if

total = rate * nights.

if nights > 7 then

discount = total * 0.15

final Price = total - discount.

Else.

final Price = total -

End if

total revenue + final price.

display final price, total revenue.

End.

Question 2 :-

Date:

Elevator Simulation

Algorithm

Start

initialize current floor = 0

input requested floor.

if requested floor > current floor
then display "moving up."

Else if

requested floor < current floor
then display "moving down"

Else display " doors opening "

current floor = requested floor

Display current floor after stop

End if

End.

PAC-

input	Processing / output.
current floor = 0 (start)	Processing -
requested floor (N)	compare requested vs current -
	Output:
	moving up / moving down / doors opening
	current floor update

IPO chart.

Date: _____

Input.	Process	Output.
current floor(0)	compare requested floor with current floor using if-else.	moving up/ moving down/ doors opening.
Requested floor	update current floor = requested floor	current floor after stop.

Pseudocode.

Begin

Set current floor = 0

Read requested floor

If requested floor > current floor then

Display "moving up"

Else if requested floor < current floor then display "moving down."

Else.

display doors "opening"

End if

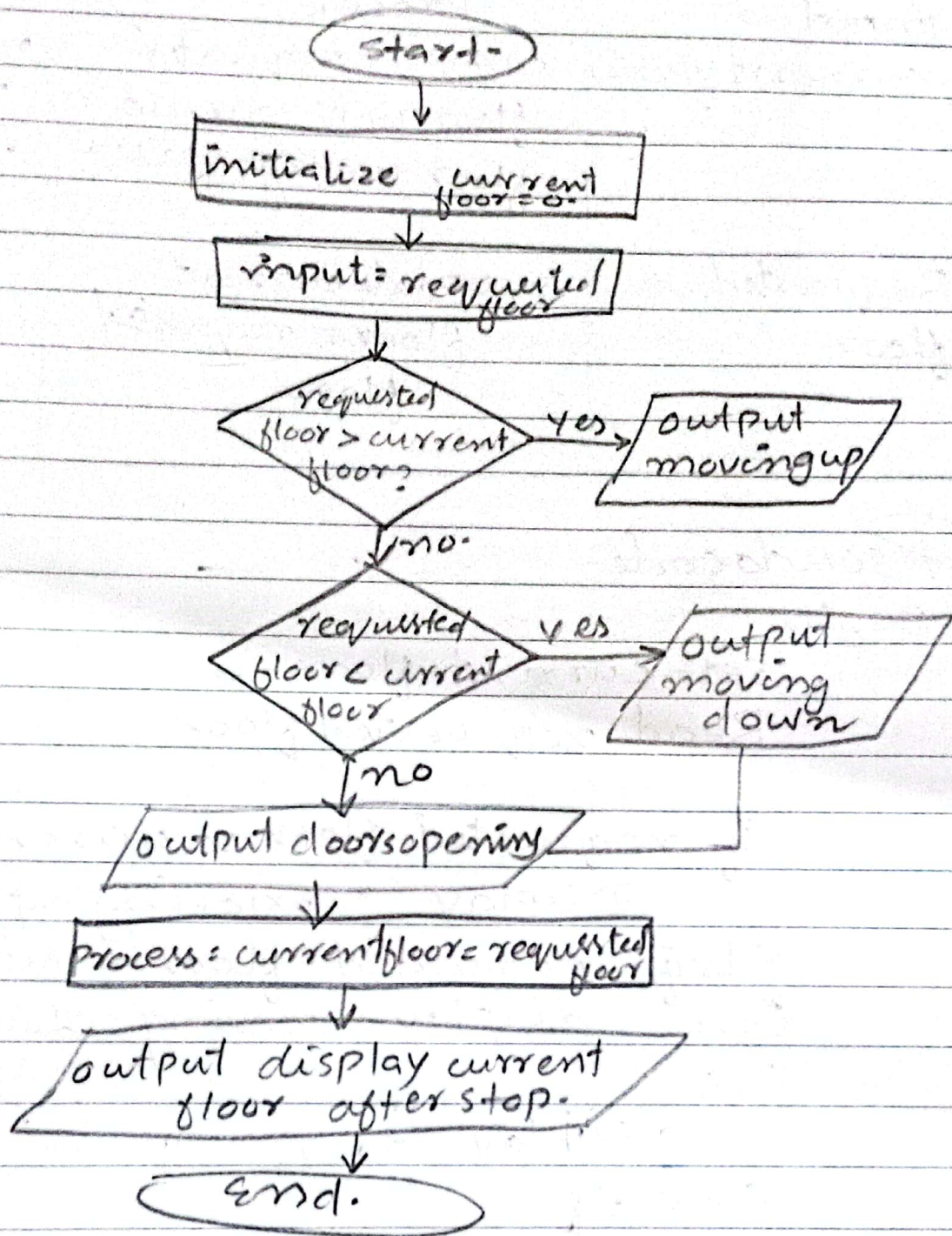
Set current floor = requested floor

Display "current floor after stop ="
current floor

End.

Flow chart: Elevator simulation

Date: _____



Question 3:-

Date:

Class result processing.

Algorithm.

- 1 Start
- 2 input m_1, m_2, m_3, m_4, m_5 .
- 3 $sum = m_1 + m_2 + m_3 + m_4 + m_5$.
- 4 $average = sum / 5$
- 5 if any mark < 33 then
result = fail - subject deficiency
- 6 Else if $average \geq 80$ then
result = "distinction"
- 7 Else if $average \geq 60$ then
result = "Pass"
- 8 Else result = fail
- 9 End if
- 10 Display sum, average, result.
- 11 End.

PAC

input	Processing / output.
5 subject marks (m_1 to m_5).	Processing = sum = total, $average = sum / 5$, check < 33 } check $average \geq 80, > 60$).
	Out put: sum, average, result, (distinction / Pass / fail / subject deficiency

TPO Chart.

Date: _____

input	process	output
5 marks (int).	sum: $m_1 + m_2 + m_3 + m_4 + m_5$ $avg = sum / 5$ if-else ladder for deficiency and classification.	sum, average final result.

Pseudocode.

Begin

Read m_1, m_2, m_3, m_4, m_5 .

Sum = $m_1 + m_2 + m_3 + m_4 + m_5$.

$avg = sum / 5$

if $m_1 < 33$ or $m_2 < 33$ or $m_3 < 33$

or $m_4 < 33$ or $m_5 < 33$ then

result = "fail - subject deficiency"

Else if $avg \geq 80$ then

result = "distinction"

Else if $avg \geq 60$ then

result = "Pass"

Else

result = "fail"

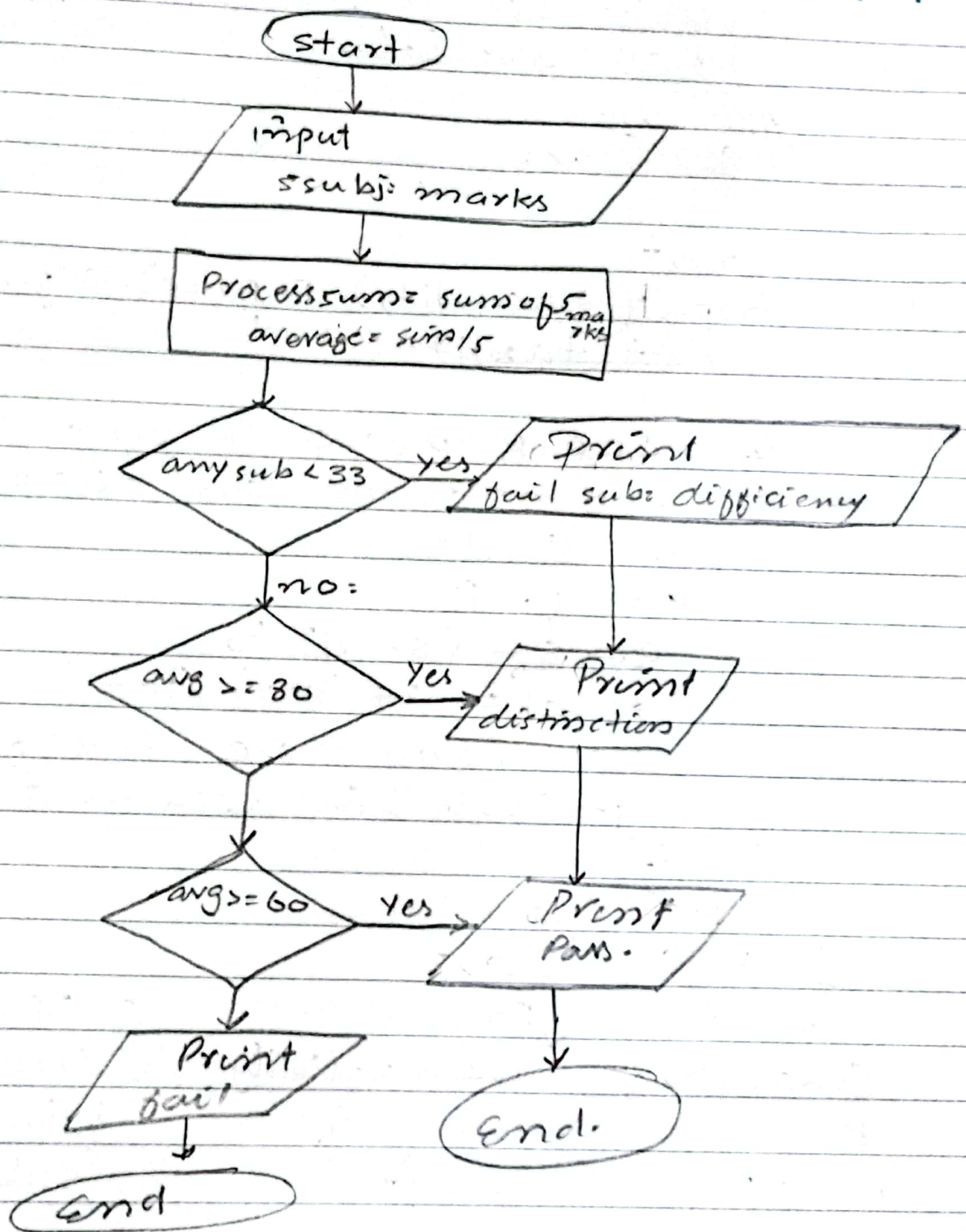
End if

Display sum, average, result.

End.

Flow chart:

Date: _____



Question 4:- Online shopping Bill Calculator.

Algorithm

Start

Input q (quantity), P (price), d (for discount %), t (tax %).

If $q \leq 0$ OR $P \leq 0$ OR $d \leq 0$ OR $d > 100$ OR $t > 0$ OR $t > 100$ then display error and End.

$$S = q * P$$

$$a = S - (S * d) / 100$$

$$\text{final} = a + (a * t) / 100$$

display and store S , a , final
generate bill document

End if

End.

I PO Chart.

Input	Process	Output.
Quantity q	check validity	Error message
price P .	using if	if invalid.
	$S = q * P$	Subtotal S .
Discoun d	$a = S - (S * d) / 100$	discounted amount
tax t	$\text{final} = a + (a * t) / 100$	final Bill

Problem.	Calculate final shopping bill in 3 dependant stages.
Input	q, p, d, t .
Process	Validate inputs, $s = q - p$, $a = s - (s * d) / 100$ final = $a + (a * t) / 100$
Output.	Subtotal, discounted amount final Bill + bill + tax.

Pseudo code.

Start.

Read ~~q~~ p, q, d, t .

If $q < 0$ OR $p < 0$ OR $d < 0$ OR $d > 100$
OR $t < 0$, OR $t > 100$.

Print "invalid input"

Stop.

Endif

$s = q * p$

$a = s - (s * d) / 100$

final = $a + (a * t) / 100$

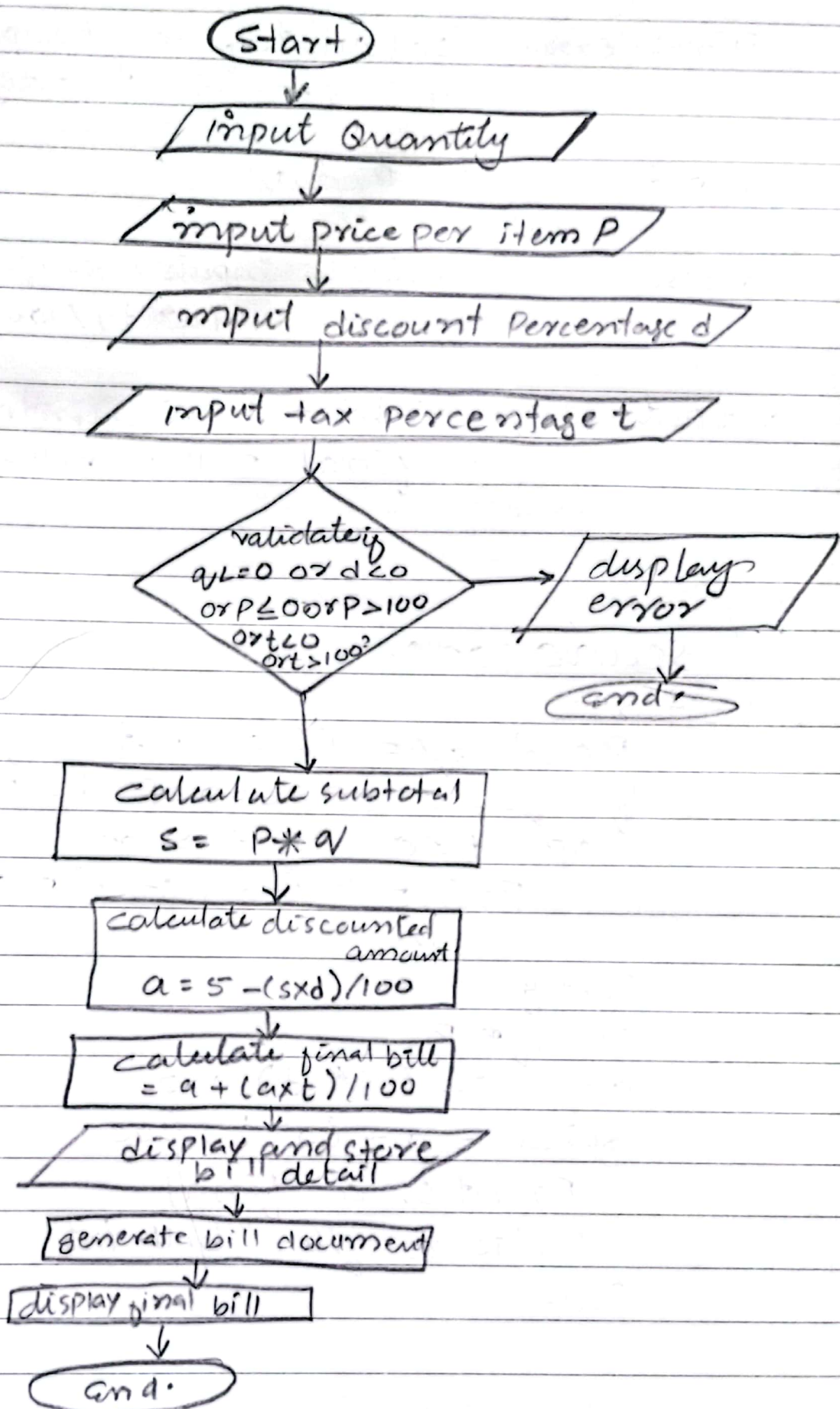
Print s, a, final

Write to file "bill.txt"

End.

Flow chart:-

Date:



Question 5:-

Date: _____

Smart campus parking and access management system.

Algorithm

Start

Initialize capacities - Zone A = 20, Zone B = 40, Zone C = 15 and current occupancies

input vehicle type (C = car, B = Bike, V = van)

input user category (F = faculty, S = student, G = Visitor).

input permit Y / N

if permit == N then input emergency = Y / N

if emergency == N then reject and end.

endif

End.

if user == S then

if vehicle == V then if zone C has 2 spaces then assign C Else reject - Else if zone C has space then assign C Else reject.

display assigned zone and remaining capacity

End if

End -

PAC

Date: _____

~~Field~~ Component description

problem definition * To design a system that checks parking eligibility of a single vehicle and assigns it to ~~the~~ correct parking zone based on user category, vehicle type, and permit status.

inputs Required.

- * Vehicle type - (C = car, B = Bike, V = Van).
- * User category (F = faculty, S = student, G = visitor)
- * Permit status (Y/N)
- * Emergency status (Y/N) if permit is invalid.
- * Current occupancy of zone.

Logic / Processing

- * validate permit if no check emergency
- * use switch for user category F/S/G
- * use nested if vehicle type (V needs special handling)
- * check capacity zone
- * A = 20, Zone B = 40
- * Zone C = 15 van in zone C requires 2 spaces.
- * Apply rejection rules if zone is full
→ continue.

Date: _____

Outputs:

- * Accept ID with assigned zone name (A/B/C).
- * rejected with reason (invalid)
- * Permit / zone full / insufficient space)
- * remaining capacity of the assigned zone.

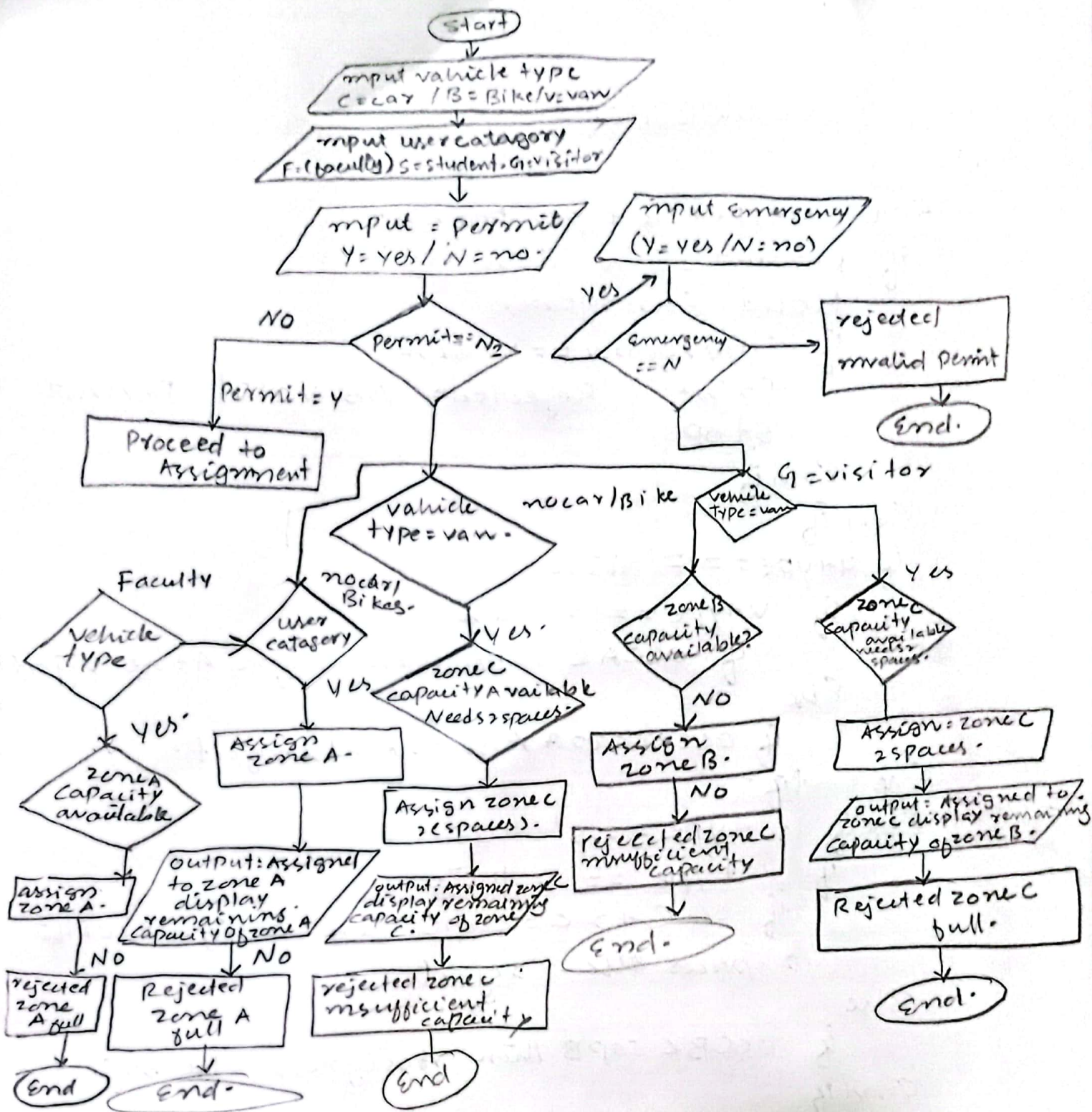
Constraints:

- * Faculty can only use zone A.
- * Student (car / Bike) can only use zone B.
- * Student van is redirected to zone C and needs 2 spaces.
- * Visitor (all types) - must use zone C.
- * No loops allowed, single transaction only.

IPO Chart-

Date:

Input	Process	Output
Vehicle type C/B/V	input validation check if v type is valid using if-else if invalid show error.	Error message: "invalid vehicle type"
User Category	Permit validation if permit == 'N' then check emergency if emergency == N then terminate.	Rejected = invalid Permit.
Permit Y/N	Assignment discision Use switch(u type) to determine user. inside each case, use nested if (v type == v') to check if van needs 2 spaces.	Assigned zone A, B or C.
Emergency Y/N Conditions:-	capacity checking if $occA < capA$ for faculty if $occB < capB$ for student if $occC + 2 < capC$ for van if $occC < capC$ for visitor.	Accepted + remaining capacity or rejected : zone full.



Pseudocode.

Date:

Start

Read v type, u type, permit

if permit == 'N' then

Read Emergency

if Emergency == N then

Print " Rejected= No valid permit"

Stop

End if

End if

if u type == 'F' then

if v type == v then

if OCCA < capA then assign A else reject

Else

if OCCA < capA Assign A Else reject -

~~End if~~ End if -

Else if u type == 'S' then

if v type == v then

if OCC + 2 <= capC then assign C with
2 spaces else . reject .

Else

if OCCB < capB then Assign B else reject -

End if

Else if u type == 'G' then

if v type == v then

if OCC + 2 <= capC then assign C else reject -

Else

if OCC < capC then assign C Else reject .

End if

End if

Print Assigned zone + Remaining -

End.

Question 6:-

Date: _____

Smart EV charging and Parking management system.

Algorithm

Start

Input vehicle type, SOC, Required level, duration, current time, membership, disabled status, station available

if station available == N then

if vehicle type == 'H' then "display charging unavailable
Parking only"

Else display "no charging slot available".

Goto End.

if vehicle type == H and SOC >= 40 then

Display "vehicle does not qualify for
EV charging."

Goto End.

needed = required level - SOC

if needed <= 0 then display "no charging required" Goto
End.

if SOC <= 15 And required level >= 80 then priority =
"emergency charging priority"

Else if disabled status == Y OR membership == Y
And SOC <= 30 then

priority = "priority charging"

Else priority = "normal charging"

if current time >= 17 and current time < 22 then

Rate = 50, Peak status = "peak"

Else rate = 35, Peak status = "off peak"

```

charging cost = needed * rate
if priority = Emergency and membership == y then
    if rate == 35 then discount C = charging cost *
    0.20 Else discount C = charging cost * 0.10
Else discount C = 0
Final charging = charging cost - discount C
if duration <= 2 then parking cost = 200
Else if duration <= 5 then parking cost = 400
Else parking cost = 700
if disabled status == 'Y' then final parking = 0,
discount P = parking cost.
Else if membership == y then discount P = parking
cost * 0.20, final parking = parking cost - discount P
Else discount P = 0, final parking = parking cost.
Final Payable = Final charging + final parking.
// Discount
final charging + final parking = final Payable.
if duration > 8 then msg = "Log - stay warning:
Please relocate your vehicle after charging."
Else msg = "stand and parking duration."
display vehicle type, soc, required level, rate
priority, Peak status, charging cost.
parking cost, discounts final payable, msg
End.
    
```

PAC.

Date:

Component.	Description.
Problem definition.	determine admission, charging eligibility - Priority, and total cost for EV/Hybrid at mall.
Inputs.	Vehicle types (E/H), SOC, (%) required level (%), duration (hrs), current time (0-23), membership (Y/N), disabled (Y/N), station available (Y/N).
Processing	check station, check hybrid eligibility ($< 40\%$), calculate needed, assign priority in order, determine peak/off-peak, calculate charging cost with discount rule, calculate parking cost with member / disabled rule check long-stay.
Output.	Vehicle type, SOC required %, priority, peak / off-peak, charging cost, parking cost, discounts, final payable warning messages.
conditions	Station N, hybrid $SOC \geq 40\%$ needed ≤ 0 , $SOC \leq 15$ and $req \geq 80$ disabled = Y member = Y and $SOC \leq 30\%$ time (17-22) duration > 2 , > 5 , > 8

I PO chart-

Date:

Input	Process	Output
1. vehicle type	1- needed = require	1. charging
2. soc	d - soc	Priority
3. required Level	2- check station availability	2- Peak status
4. duration	3- check hybrid condition	3/ charging cost & discount
5. current time	4- priority logic using nested if-else.	4- Parking & discount.
6. membership	5- rate = 35/50 based on time	5- final payable amount
7. disabled	6- charging cost = needed rate - discount.	6- duration warning
8. Station available	7- Parking cost based on duration & slab discounts	

+ pseudocode.

Begin.

Date:

Read vehicle type, soc, required, duration, time, member, disabled, station

if station == 'N' then

if vehicle type == 'H' Then print "

"Charging unavailable - Parking only."

Else print "no charging slot available"

Exit.

End if

if vehicle type == 'H' and SOC >= 40 then

Print "vehicle does not qualify for EV charging."

Exit

End if

needed = required - soc

if needed <= 0 then

Print "no charging required"

Exit

End if.

if SOC <= 15 and required >= 80 then priority = "Emergency charging priority"

Else if disabled == 'Y' or (member == 'Y' and SOC <= 30) then priority = "Priority charging"

Else priority = "normal charging"

if time >= 17 and time < 22 then rate = 50
status = "peak"

Else rate = 35; status = "off-peak"

Date:

charge = needed * rate.

if priority == "Emergency charging
Priority" then discC = 0

Else if if member == 'Y' then if rate == 35
then disc C = charge * 0.2 Else disc C =
charge * 0.1

if duration <= 1 then park = 200 Else if duration > 5
then park = 400 Else park = 700

if disabled == 'Y' then disc P = Park;
final park = 0

Else if member == 'Y' then disc P = Park * 0.2;
Final park = Park - DiscP

Final charge = charge - DiscC.

final = final charge + final park

if duration > 8 then Msg = "Long-stay-
warning----"

Else Msg = "Standard parking duration."
Print all values.

End. if
End.

